BILLA BALA SATYA PHANEENDRA

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Education

Koneru Lakshmaiah Education Foundation — Vaddeswaram
Bachelor of Technology in Electronics and Communication Engineering — June 2023 - Present
| CGPA: 8.65 / 10.00

Rajadhani Jr Collage — Guntur
SSC, Class X | %: 81 / 100 — June 2021 – March 2023

Sri Chaitanya Techno School — Guntur
SSC, Class X | %: 99 / 100 — June 2020 – March 2021

Skills

Interests: Verilog· RTL Design ·Drones · Embedded Systems· DSP ·Physical Design

Programming: C language· Java· Verilog ·System Verilog ·UVM

Design Software: Xilinx Vivado · Cadence Virtuoso ·LTSpice ·Arduino IDE ·MATLAB ·Synopsys Design Compiler

Skill: AI Prompt Engineering

Projects

Integrated V2X and Multi-Band Beamforming Antenna for Autonomous Vehicles — April 2026 - Present

- Developed a conformal multi-band high-gain beamforming antenna system for V2X communication and K/Ka-band radar applications in autonomous vehicles
- Implemented high-gain beamforming techniques to enhance vehicular communication, radar sensing, and intelligent transportation system performance

URAT Synthesis using Vivado — February 2026

- I have analyzed post-synthesis reports, used checkpoints for iterative analysis, and observed the impact of different synthesis settings on performance.

CMOS S-Parameter based Low Noise Amplifier — OCT 2025 – NOV 2025

- Designed and simulated a 2.4 GHz CMOS Low Noise Amplifier using S-parameter analysis, achieving a gain below 15 dB.

Autonomous Quadcopter Drone — Jan 2025 – April 2025

- Implemented autonomous features like real-time monitoring for performance analysis and optimization
- Presented the project in Electronic System Design Skill Development Course Review (May 2025)

Home Automation System using the STM32 — September 2024

- Executed a Bluetooth-controlled LED automation prototype using the STM32 Blue Pill and HC-05 module, enabling wireless ON/OFF Control of three LEDs through a mobile application.

Detection Based Security System — May 2024 – July 2024

- Designed and implemented a real-time Security System using an ESP8266 Node MCU.

Achievements

IEEE AP-S Undergraduate Research Funding Award – 2026

• Awarded a USD 3,000 research grant by the IEEE Antennas and Propagation Society (IEEE AP-S) under the IEEE AP-S Undergraduate Research Funding Program 2026.
• Selected for the project: "A Conformal Integrated-Aperture Multi-Band Antenna System for Intelligent Vehicles: V2X (5.9 GHz) and High- Gain Beamforming at K/Ka-Bands (24/28/39 GHz) for Radar-Communication in Autonomous Vehicles."

Research Publication

Artificial Intelligence-powered Drone for Wildlife Monitoring and Animal Detection

Accepted for presentation at WAMS 2026 Conference (Paper ID: 17)

Internships

AICTE Virtual Internship by Microchip
Embedded System Developer — 04/2025 - 06/2025

Skill Dzire
VLSI Internship Program — 05/2025 - 06/2025

Hackathons

SDG Innovation Challenge -Techno Summit'25
September 2025
- Sathyabama University
- Secured 4th place at the SDG Innovation Challenge.
- Developed NP-Scan for Smart Pesticide Screening aligned with SDG3.
  Research Existing detection methods and key challenges.

Certification

Cambridge Linguaskill Test (Achieved B1 level)
Cambridge — July 2024

Artificial Intelligence and Machine Learning Using Python
September 2024
Taras Systems and Solutions

Altium Designer Essentials: Mastering the Fundamental
January 2025
Altium

Embedded System Application and IOT Programming
April 2025
Taras Systems and Solutions

Spectre Simulator Fundamentals S1
April 2025
Cadence

VLSI Design and Verification – L1
December 2025
Taras Systems and Solutions

System Verilog & UVM -L2
March 2026
Taras Systems and Solutions

Positions of Responsibility

Vice President, PULSE
ECE Student Body KL University — April 2026 - Present

Chair, IEEE Antennas and Propagation Society (AP'S)
January 2026 - Present
Student Branch KL University

KL University Alumni Relations
April 2026 - Present
Chief Student Coordinator

NSS Core Unit-8 KLEF
January 2026 – present
Campus relations

KL researcher's Club
September 2024 – October 2025
Financial Lead

Extra Circulars Activities

Organized Samyak -25
October 2025
- Headed Sponsorship and Partnership operations as Chief.
- Coordinated with marketing and event sponsor visibility.

Organized Surabhi - 26
March 2026
- Organized guest accommodation, transportation as Lead Hospitality
- Resolved on-ground issues efficiently, improving guest experience.